

4th International Conference on Innovative Data Communication Technology and Application

American Sign Language Recognition for Alphabets Using MediaPipe and LSTM

Sundar B^a, Bagyammal T^b

^aAmrita school of Computing, Amrita Vishwa vidyapeetham, Coimbatore- 641 112., India

^bAmrita school of Computing, Amrita Vishwa vidyapeetham, Coimbatore- 641 112., India

Abstract

With the advancement of today's technologies in artificial intelligence, humans tend to use hand gestures in their communication to convey their ideas. Gesture recognition is an active area of research in the human-computer interface (HCI). Gesture recognition is important for communication between deaf-mute people, HCI, robot control, home automation, and medical applications. In this article, a simple and efficient vision-based approach for American Sign Language (ASL) alphabets recognition has been discussed to recognize both static and dynamic gestures. Mediapipe introduced by Google had been used to get hand landmarks and a custom dataset has been created and used for the experimental study. Hand gesture recognition has been done by using Long short-term memory (LSTM). The proposed system has been investigated with 26 alphabets and an accuracy of 99% has been achieved. This work can be used to convert hand gestures into text.

© 2023 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 4th International Conference on Innovative Data Communication Technologies and Application

Keywords: Hand Gesture Recognition, American Sign Language (ASL), Mediapipe, Long Short Term Memory (LSTM), Human-Computer Interface.

1. Introduction

The advancement in the development of various technologies for smartphones, laptops, notebooks, and various handheld devices like game applications flourishes research in the gesture recognition domain. Initially, a wearable hand gloves-based sensor was used for the gesture recognition system, but the production cost is high. Now camera vision-based sensors are used because it is user-friendly and simple. Gestures are any particular patterns or motions of the hands, face, or body used to create certain expressions in sign language. Additionally, static hand positions and

* Corresponding author. Tel.: +91-9677682940.

E-mail address: t.bagyammal@cb.amrita.edu, cb.en.p2aid20041@cb.students.amrita.edu

configurations without any movement can be used to express gestures. A single hand position can represent a stance, whereas multiple hand positions can be combined to represent a continuous motion [1].

The proposed system uses ASL for alphabet recognition. ASL is one of the most widely used languages in the world. There are around 250,000 to 500,000 people using sign language on the daily basis around the world [6]. It is very difficult for people to understand sign language if they are not familiar with it. Gesture recognition can be used in these situations to translate gestures into natural language text for communication.

The aim of the study is to help people recognize alphabets that are communicated using ASL. The proposed system can be used to translate the sign language into text and HCI. The novelty of the proposed system is that it uses Google's MediaPipe for hand landmark detection which is more accurate and faster when compared to the conventional methods that use geometric, shape, and edge features. LSTM model was found to be very effective for sequence data modeling and has been used for gesture recognition.

2. Literature Survey

Many authors have proposed various image processing and Machine Learning (ML) techniques for sign language recognition. Some of the relevant works have been discussed here.

In the middle of the 1970s, Myron W. Krueger originally suggested gesture recognition as a brand-new method of communication between people and computers. With the rapid advancement of computer hardware and vision systems over the last few years, it has emerged as a very significant study topic. A. A. Abdulhussein and F. A. Raheem [1] discussed how to use grayscale images and edge detection techniques to detect hand gestures. The drawback of this technique is that there are some limitations when it comes to grayscale images as they are only 2D data and it is difficult to extract the key features of the hand. Teak-Wei Chong and Boon-Giin Lee [2] introduced a technique to recognize hand gestures using a leap motion controller which is quite expensive and requires additional hardware. Matteo Rinalduzzi [3] introduced a technique that uses gloves and several hand sensors to recognize different hand gestures. The sensors have motion detectors that are costly, and it makes the hand hard to move because of the weight. Rajesh George Rajan and Dr.M.Judith Leo [4] discussed a technique using deep learning to recognize the ASL gestures with the help of a public dataset available on Kaggle. It employs several ML algorithms in the dataset and gives a classified report on the results for each ML algorithm.

Aashni Hariiaa, et al [12] used color features and contour extraction to determine the hand gestures. The drawback here is that since it uses contour extraction it can be used for counting the number of fingers shown rather than the different hand gestures. Pei Xu [14] discussed an approach for realtime hand gesture recognition for human computer interaction using CNN. The drawback here is that it does not detect any sign language gestures rather it detects some hand signs that are used for Human computer Interface rather than a sign language. Abhishek Bet al [16] proposed an algorithm to detect hand gestures using 3D CNN. However, how to detect gestures with motion has not been discussed.

Gongfa Li, et al [17] explained a method to detect hand gestures using CNN with the help of Kinect. The drawback is that Kinect is quite expensive. Noorkholis Luthfil Hakim, et al [18] discussed an approach for recognizing dynamic hand gestures using 3DCNN and LSTM. The drawback here is that they don't recognize gestures for alphabet but rather use them for military gestures.

Sourav Bhowmick, Sushant Kumar, and Anurag Kumar [5] introduced a technique to recognize ASL gestures using Deep Learning for two-letter combinations. The drawback of this method is that this technique uses still images and it will be difficult to track gestures with a motion. Gestures that contain motion are not recognized accurately. Onamon Pinsanoh, Yuttana Kitjaidure, Ariya Thongtawee [7] used a technique to find the gestures by dividing the frame into left and right sides. But this process increases the complexity of gesture feature extraction. Chinmaya R. Naguri, et al [8] introduced a technique to detect dynamic hand gestures from 3D motion. This system works using a Leap motion controller. Kai Li, Qieshi Zhang, Jun Cheng, Jianming Liu [9] illustrated how to use the HCI using hand tracking with hand gesture recognition using hand region segmentation. They used it for hand tracking and combined it with hand gesture recognition. Thakur, et al [25] illustrated on real time sign language recognition and speech generation. They used CNN algorithm in the paper. The drawback here is they have replaced the signs that have motion with static images. It is observed from the background study that research on gesture recognition is still an active area of research and new methods are required to effectively make use of the diverse and new computer vision algorithms.

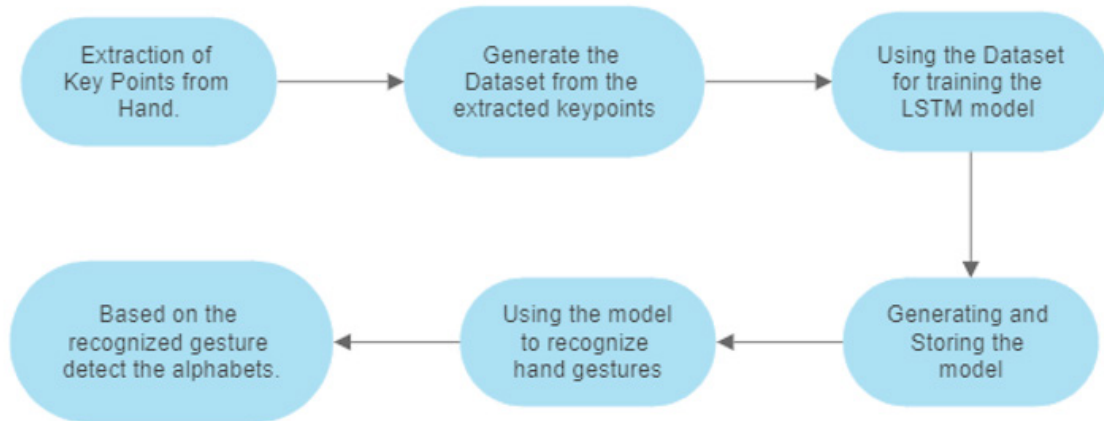


Fig. 1. Architecture diagram of the system.

3. Proposed System

The proposed system for the ASL alphabet recognition is shown in Figure 1. The proposed system recognizes the American sign language of alphabets. It recognizes gestures with motion and gestures without motion as well. The first step is to collect the dataset for the system which will be discussed in results and discussion section. MediaPipe hands is a high- fidelity hand and finger tracking framework used in the dataset preparation. It makes use of a number of ML models to infer the 21 three dimensional landmarks of a hand in real time from just one frame.

3.1. Mediapipe

Mediapipe hand key point detection works with a machine learning pipeline. A palm detection model that works on the entire image is one of the models in the MediaPipe machine learning pipeline. It receives the entire image and outputs a hand-bounding box that is orientated. a hand landmark model that uses the palm detector-cropped image region and outputs highly accurate 3D hand keypoints[15]. Additionally, the hand is cropped by the ML pipeline using the hand landmarks found in the previous frame, and palm detection is only used to localise the hand when the landmark model is unable to do so. Figure 2 shows the 21 hand landmarks that can be tracked from the Mediapipe's hand landmark detector. Arsheldy Alvin, et al [13] have used Mediapipe and K nearest neighbours algorithm to determine the hand gestures which helps us in understanding how to use mediapipe for hand gesture recognition.

The difference between tensorflow and mediapipe is that tensorflow is a free and open-source software library for machine learning and artificial intelligence. It can be used across a range of tasks but has a particular focus on training and inference of deep neural networks. Whereas Mediapipe offers open source cross-platform, customizable ML solutions for live and streaming media. In this paper we have used mediapipe for capturing the hand keypoints and tensorflow for training and detecting the ML algorithm. Mediapipe runs with both on CPU and GPU. No additional processing power is required to run Mediapipe.

The dataset required for the experimental study of the proposed algorithm has been created using mediapipe open source library. The mediapipe hand tracking library has around 21 key features in hand and it is used to generate the dataset. The generated dataset contains all the 26 alphabets with hand gestures. It captures the hand gestures for 30 frames. The dataset has 30 different permutations for each alphabet. The same dataset is collected from four different humans including 2 males 1 female and 1 kid. As a total the dataset contains 93,600 values. The dataset has different parameters that are time, values(x,y,z) of hand joints.

3.2. Sign Language Recognition Using LSTM

The next step is to train the LSTM model using the dataset. LSTM is one of the feed- forward neural networks. Since, in American alphabet hand gestures there are several hand gestures that have hand motion in them instead of a

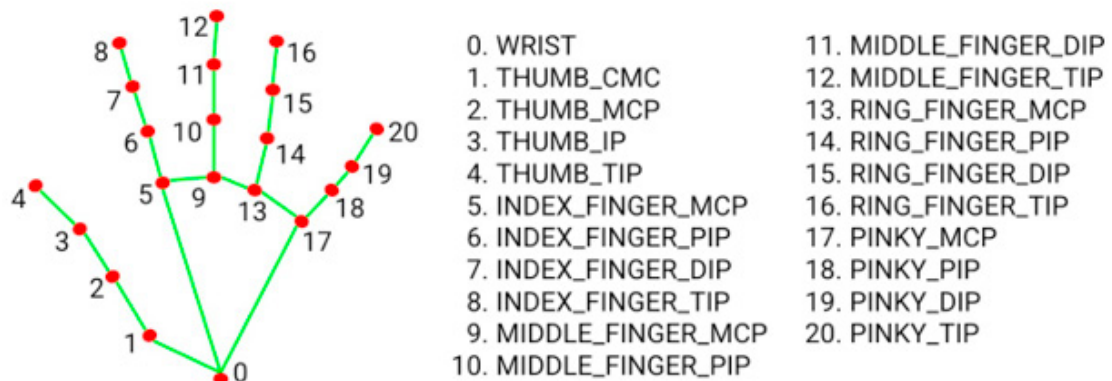


Fig. 2. Hand Landmarks representation used in Mediapipe[?].

single static pose. So, its better to use LSTM using which we can recognize the continuous motion. The advantage of using LSTM is that it allows not only single data points i.e images, but it also allows the entire sequences of data[19]. An artificial neural network is a network of interconnected neurons, each of which is connected to one another and is modelled after the brain's organic neural networks. We can execute sophisticated operations on the data by combining a number of different algorithms, each of which is capable of completing a different task [20]. A class of neural networks called Recurring Neural Networks, or RNNs, is designed specifically to handle temporal data. The input is processed using the internal state of the RNN neurons, which is accomplished with the aid of loops built into the neural network. RNN neurons have a cell state or memory.

From the training dataset, Spatio-temporall features are extracted and the LSTM model classification model is derived. The generated ML model is stored for prediction of gestures in real time. All the 26 alphabet gestures are collected one after the other and are preprocessed into numpy arrays and stored. The dataset collected is then passed onto the LSTM model. The LSTM model will have all the 26 alphabet classes for each of the alphabets. The algorithm is made to run on 2000 epochs until the accuracy becomes closer to 1 and training loss goes to nearly 0. The algorithm runs with the above specified features and generates the model which is stored for future use. The model can now be used for tracking the live hand gestures and for further operations.

The system uses RELU activation. The rectified linear activation function, also known as ReLU, is a piecewise linear function that, if the input is positive, outputs the input directly; if not, it outputs zero. Because a model that utilises it is simpler to train and frequently performs better, it has evolved into the default activation function for many types of neural networks[21].

The system uses Adam Optimizer technique. In place of the conventional stochastic gradient descent method, Adam is an optimization technique that may be used to repeatedly update network weights depending on training data [10]. For all weight updates, stochastic gradient descent maintains a constant learning rate (referred to as alpha), which does not change throughout training. As learning progresses, a learning rate is maintained and independently adjusted for each network weight (parameter).The main reason for using the LSTM model is that we can detect motion from hand gestures, and it can be used to detect hand gestures that have motion in them.

In Figure 3, we can see the diagram of LSTM cell. The LSTM cell has three steps namely the forget gate, input gate and output gate. The forget gate establishes how much the prior data should be forgotten. The amount of information to be printed on the internal cell is determined by the input gate. The output gate decides what output to produce based on the internal cell state at the moment. Electric energy consumption modelling using LSTM has been studied in [22]. LSTM based natural language processing model has been used for Malayalam language analysis[22],[23] and paraphrase identification[24].

The model that we are using is sequential and they are faster to train and have faster detections. The network implemented here consists of six layers three LSTM layers and three dense layers. In Figure 4 you can find the summary of the model with the layer type , output shape and params.

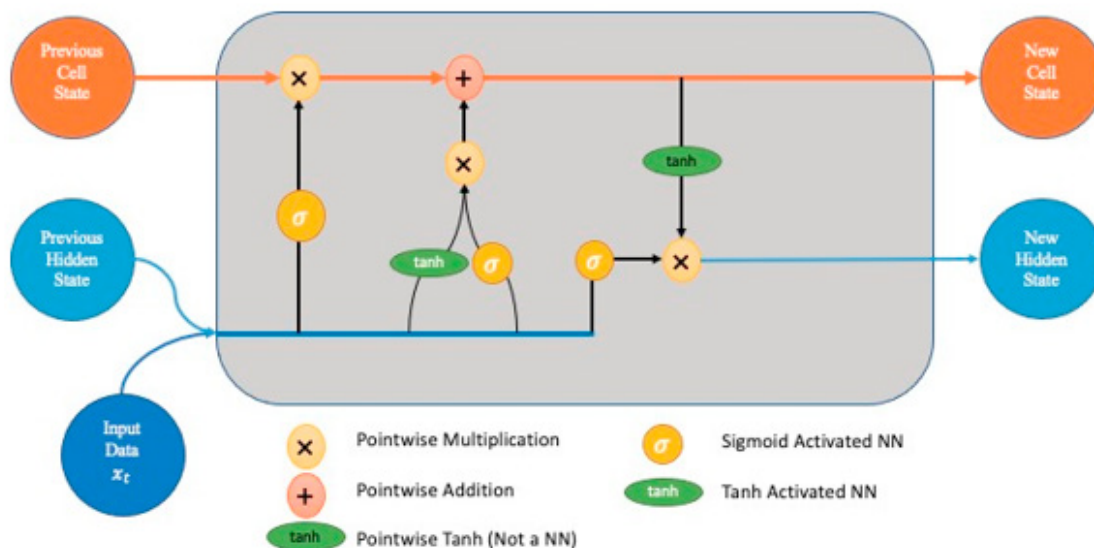


Fig. 3. LSTM Architecture.

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 30, 64)	32768
lstm_1 (LSTM)	(None, 30, 128)	98816
lstm_2 (LSTM)	(None, 64)	49408
dense (Dense)	(None, 64)	4160
dense_1 (Dense)	(None, 32)	2080
dense_2 (Dense)	(None, 26)	858
=====		
Total params: 188,090		
Trainable params: 188,090		
Non-trainable params: 0		

Fig. 4. Summary of the sequential model.



Fig. 5. Gestures for Each Alphabets from ASL[11].

4. Results and discussion

4.1. Dataset

The dataset required for the experimental study of the proposed algorithm is created using MediaPipe which is an open-source library. It captures the hand gestures for 30 frames and stores them as NumPy array for training the model. The dataset has 30 different permutations of a single alphabet. The same process is repeated with four different demographics of humans to get a diverse dataset. The four different humans are two male hands, one female hand, and one kid's hand. This makes a total of 93,600 numpy array, these are indexed and stored. The dataset has different parameters that are x, y and z axes of the hand joints. The dataset is divided into 90% for training and 10% for testing. In Figure 5, the ASL hand gestures of all the alphabets from A to Z used in the proposed system are shown.

4.2. Experimental results

Table 1 shows the sample image result of gesture recognition for the alphabet from A to Z. Dots connected with lines in hand regions denote the hand landmarks.

Some characters have similar gestures. So, it's difficult to identify the gestures correctly. Some of the examples of false detections are shown in Figure 6. All these characters have a similar hand gestures and so it's difficult to recognize the correct gestures. After altering the gestures accordingly and making improvements in the gestures these false detections are avoided.

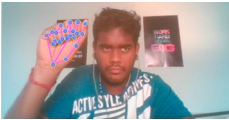
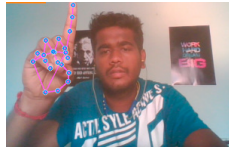
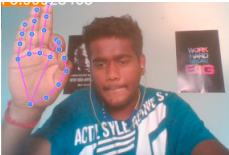
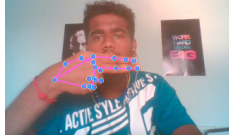
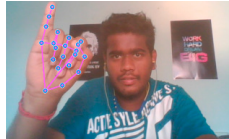
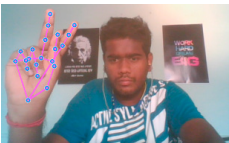
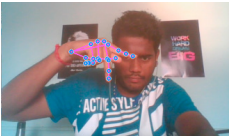
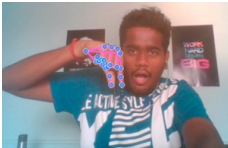
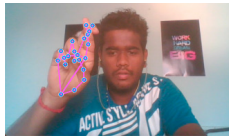
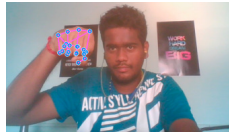
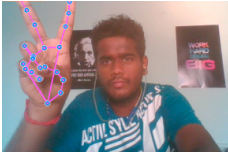
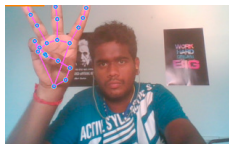
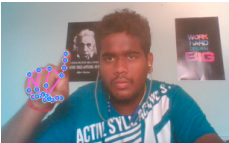
Figure 6 shows some gestures that are misrecognized. The hand gesture for alphabet V is recognized as U and the hand gesture for alphabet A is recognized as S. These recognitions are false detections and are corrected after making the improvements in the form of increasing the dataset and altering the gestures in a way so that the system can recognize it.

After making the changes in the above dataset the model is retrained with increased dataset and with improved gestures for some hand gestures. This process is done to reduce the recognition errors.

The result of alphabet recognition is shown in Figure 7 as a confusion matrix. It can be inferred from the confusion matrix that most of the values are predicted accurately and there are some values that are not predicted accurately. Alphabets like R, and V have some false predictions in them. The predictive error for all the inputs is obtained and is tabulated in the confusion matrix.

Various evaluation metrics of the experiment are given in Table 2. Evaluation metrics used are precision, recall, F1-score, and support. It can be understood that most of the alphabets are recognized with good precision, recall, and F1-score. Alphabets that are difficult to be recognized are L, M, Q, R, U, V, W whereas other rest of the alphabets have been recognized correctly. Evaluation metric values are calculated based on the confusion matrix content depicted in Figure 7 and explained using the equations from equation 1 to equation 3.

Table 1. Screenshots of the Results.

A	B	C	D	E
				
F	G	H	I	J
				
K	L	M	N	O
				
P	Q	R	S	T
				
U	V	W	X	Y
				
Z				

A true positive is an outcome where the model correctly predicts the positive class. Similarly, a true negative is an outcome where the model correctly predicts the negative class. A false positive is an outcome where the model incorrectly predicts the positive class. And a false negative is an outcome where the model incorrectly predicts the negative class.

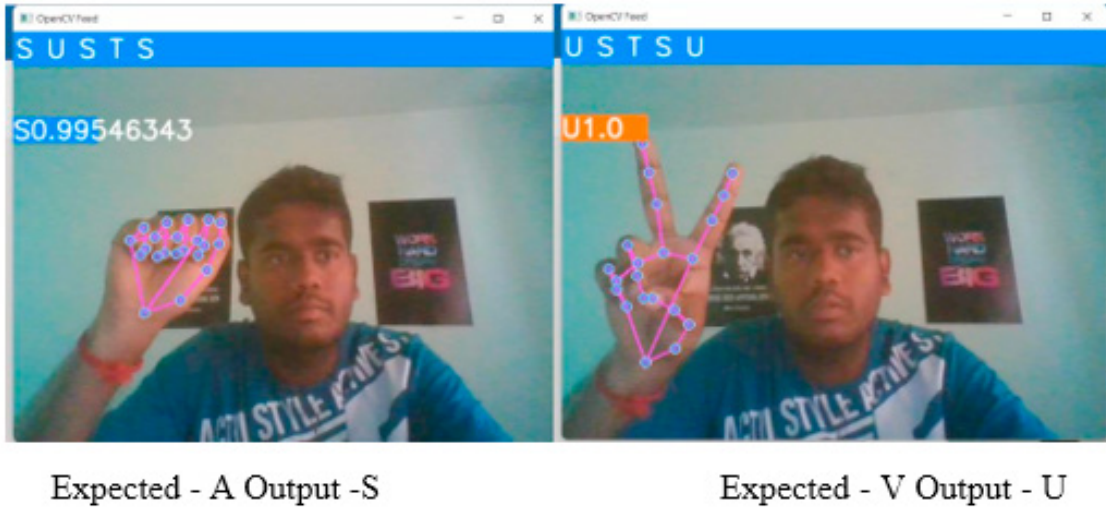


Fig. 6. False Detections of Characters.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C	0	0	13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
D	0	0	0	15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
F	0	0	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
G	0	0	0	0	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H	0	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I	0	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
J	0	0	0	0	0	0	0	0	0	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
K	0	0	0	0	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
L	0	0	0	0	0	0	0	0	0	0	0	15	0	0	0	0	0	0	0	0	0	0	0	0	0	0
M	0	0	0	0	0	0	0	0	0	0	0	0	17	0	0	0	0	0	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0	0	0	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0
O	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0	0	0	0
Q	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	0	0	0	0	0	0	0	0	0
R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	9	0	0	0	0	0	0	0	0
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0
T	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	0	0	0	0	0	0
U	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	0	0	0	0	0
V	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	10	0	0	0	0
W	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	12	0	0	0
X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0
Y	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13	0
Z	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12

Fig. 7. confusion matrix for gesture recognition of 26 letters from ASL.

The ratio of True Positives to All Positives is known as precision.

$$\text{Precision} = \frac{\text{TruePositive}(TP)}{\text{TruePositive}(TP) + \text{FalsePositive}(FP)} \quad (1)$$

The recall is the measure of the model correctly identifying True Positives.

$$\text{Recall} = \frac{\text{TruePositive}(TP)}{\text{TruePositive}(TP) + \text{FalseNegative}(FN)} \quad (2)$$

The F-score is calculated from the precision and recall of the test. The F-score or F-measure is a measure of a test's accuracy.

$$\text{F1Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

Table 2. Report of Precision, Recall, F1-score, and Support.

Alphabet	Precision	Recall	Column A	Support
A	1	1	1	13
B	1	1	1	12
C	1	1	1	13
D	1	1	1	15
E	1	1	1	8
F	1	1	1	12
G	1	1	1	16
H	1	1	1	9
I	1	1	1	10
J	1	1	1	11
K	1	1	1	9
L	0.97	1	0.97	15
M	0.94	1	0.94	17
N	1	1	1	12
O	1	1	1	10
P	1	1	1	10
Q	0.92	1	0.96	12
R	1	0.9	0.95	10
S	1	1	1	10
T	1	1	1	11
U	0.95	1	0.97	18
V	1	0.83	0.91	12
W	1	0.92	0.96	13
X	1	1	1	9
Y	1	1	1	13
Z	1	1	1	12
Accuracy			0.99	312
Micro Average	0.99	0.99	0.99	312
Weighted Average	0.99	0.99	0.99	312

The number of actual occurrences of the class in the specified dataset is Support.

5. Conclusion and Future Enhancements

The proposed vision-based system can be used to recognize the ASL alphabets and converts them into text. Medi-aPipe hand landmark features and LSTM model was found to be effective for gesture recognition. Accuracy, Micro Average and Weighted Average of the alphabets are 0.99 for the custom dataset, this corroborates that the proposed system can be effectively used for HCI. The system can be used in HCI like interacting with the computer based on the gesture performed. Furthermore example could be to use the alphabets detected on a word document to type the alphabets instead of a keyboard or for typing on the search engine. This work can be further extended to work with real time applications and multiple letters or word recognition can be done to control or interact with the applications.

References

- [1] A. A. Abdulhussein and F. A. Raheem, "Hand gesture recognition of static letters American sign language (ASL) using deep learning," *Engineering and Technology Journal*, Vol. 38, No. 06, pp. 926-937, 2020.
- [2] Teak-Wei Chong and Boon-Giin Lee, *American Sign Language Recognition Using Leap Motion Controller with Machine Learning Approach*, Department of Electronics Engineering, Keimyung University, Daegu 42601, Korea, October 2018.
- [3] Matteo Rinalduzzi, Alessio De Angelis, Francesco Santoni, Emanuele Buchicchio, Antonio Moschitta, Paolo Carbone, Paolo Bellitti, Mauro Serpelloni, *Gesture Recognition of Sign Language Alphabet Using a Magnetic Positioning System*, June 2021.
- [4] Rajesh George Rajan, Dr.M.Judith Leo, *American Sign Language Alphabets Recognition using Hand Crafted and Deep Learning Features* IEEE Xplore Part Number:CFP20F70-ART published on 2020.

- [5] Sourav Bhowmick, Sushant Kumar and Anurag Kumar, Hand Gesture Recognition of English Alphabets using Artificial Neural Network, published on 2015.
- [6] Ross E. Mitchell, Gallaudet Research Institute, Draft manuscript accepted for publication in Sign Language Studies, Volume 6, Number 3, 2006.
- [7] Onamon Pinsanoh, Yuttana Kitjaidure, Ariya Thongtawee. A Novel Feature Extraction for American Sign Language Recognition Using Webcam. Published in 2018.
- [8] Chinmaya R. Naguri, Razvan C. Bunesco. Recognition of Dynamic Hand Gestures from 3D Motion Data using LSTM and CNN architectures. Published in 2017.
- [9] Kai Li, Qieshi Zhang, Jun Cheng, Jianming Liu .Hand Gesture Tracking and Recognition based Human-Computer Interaction System and Its Applications published on 2018.
- [10] Diederik P. Kingma, Jimmy Ba. Adam: A Method for Stochastic Optimization. Published as a conference paper at the 3rd International Conference for Learning Representations, San Diego, 2015.
- [11] Clayton Valli, Ceil Lucas, Linguistics Of American Sign Language, American National Standard for Information Sciences.
- [12] Aashni Hariia , Archanasri Subramaniana , Nivedhitha Asokkumara, Shruti Poddara , Jyothi S Nayaka , Hand Gesture Recognition for Human Computer Interaction. 7th International Conference on Advances in Computing Communications, ICACC- 2017, 22- 24 August 2017, Cochin, India.
- [13] Arsheldy Alvin1, Nabila Husna Shabrina2, Aurelius Ryo3, Edgar Christian4 Fakultas Teknik dan Informatika, Universitas Multimedia Nusantara, Teknik Komputer Tangerang, Indonesia. Hand Gesture Detection for American SignLanguage using K- Nearest Neighbor with Mediapipe .
- [14] Pei Xu, Department of Electrical and Computer Engineering, University of Minnesota, Twin Cities. A Real-time Hand Gesture Recognition and Human- Computer Interaction System.
- [15] Fan Zhang ,Valentin Bazarevsky ,Andrey Vakunov ,Andrei Tkachenka ,George Sung, Chuo-Ling Chang, Matthias Grundmann. Google Research 1600 Amphitheatre Pkwy, Mountain View, CA 94043, USA.
- [16] Abhishek B, Kanya Krishi, Meghana M, Mohammed Daaniyaal, Anupama H S , BMS Institute of Technology, Bangalore, India. Hand gesture recognition using machine learning algorithms. Computer Science and Information Technologies.
- [17] Gongfa Li, · Heng Tang, · Ying Sun, · Jianyi Kong, · Guozhang Jiang, · D u Jiang · Bo Tao, ·Shuang Xu, · Honghai Liu. Hand gesture recognition based on convolution neural network. part of Springer Nature 2017.
- [18] Noorkholis Luthfil Hakim , Timothy K. Shih , Sandeli Priyanwada Kasthuri Arachchi , Wisnu Aditya , Yi-Cheng Chen and Chih-Yang Lin . Dynamic Hand Gesture Recognition Using 3DCNN and LSTM with FSM Context-Aware Model.
- [19] Sepp Hochreiter ,Fakultat fur Informatik, Technische Universitat Munchen, 80290 Munchen, Germany. LONG SHORT-TERM MEMORY, Neural Computation 9(8):1735-1780, 1997.
- [20] Deep Learning in Neural Networks: An Overview.Jurgen Schmidhuber,The Swiss AI Lab IDSIA,Istituto Dalle Molle di Studi sull'Intelligenza Artificiale,University of Lugano SUPSI,Galleria 2, 6928 Manno-Lugano,Switzerland,8 October 2014.
- [21] Deep Learning using Rectified Linear Units (ReLU). Abien Fred M. Agarap . arXiv:1803.08375v2 [cs.NE] 7 Feb 2019.
- [22] Kumar, S., M. Anand Kumar, and K. P. Soman. "Deep learning based part-of-speech tagging for Malayalam Twitter data (Special issue: deep learning techniques for natural language processing)." Journal of Intelligent Systems 28, no. 3 (2019): 423-435.
- [23] Premjith, B., K. P. Soman, and M. Anand Kumar. "A deep learning approach for Malayalam morphological analysis at character level." Procedia computer science 132 (2018): 47-54.
- [24] Aravinda Reddy, D., M. Anand Kumar, and K. P. Soman. "LSTM based paraphrase identification using combined word embedding features." In Soft computing and signal processing, pp. 385-394. Springer, Singapore, 2019.
- [25] Thakur, Amrita, Pujan Budhathoki, Sarmila Upreti, Shirish Shrestha, and Subarna Shakya. "Real Time Sign Language Recognition and Speech Generation." Journal of Innovative Image Processing 2, no. 2 (2020): 65-76.